

NPTEL

Sequential Modeling with RNNs and LSTMs

Contents

- ✓ Why Sequence Models?
- ✓ Understanding Temporal Dependencies
- ✓ Recurrent Neural Networks (RNNs)
- ✓ RNN Architecture and Computation
- ✓ Types of RNNs
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- ✓ Long Short-Term Memory (LSTM)
- ✓ LSTM Gates and Cell State

Motivation

Real-world data often has an inherent order

In NLP

➤ Sentence

➤ Speech

Example:

➤ Dog bites man

➤ Man bites dog

Same words but different meanings.

I love AI

↓
Tokenization

↓
Embedding vectors

How can we process these vectors while preserving their sequential order?

Application

- ✓ Word prediction
- ✓ Sentence classification
- ✓ Translation

SEQUENTIAL MODELING

Sequential Modeling

Concept of temporal dependency

✓ The movie is **not** good.

✓ I grew up in France. I speak fluent ____.

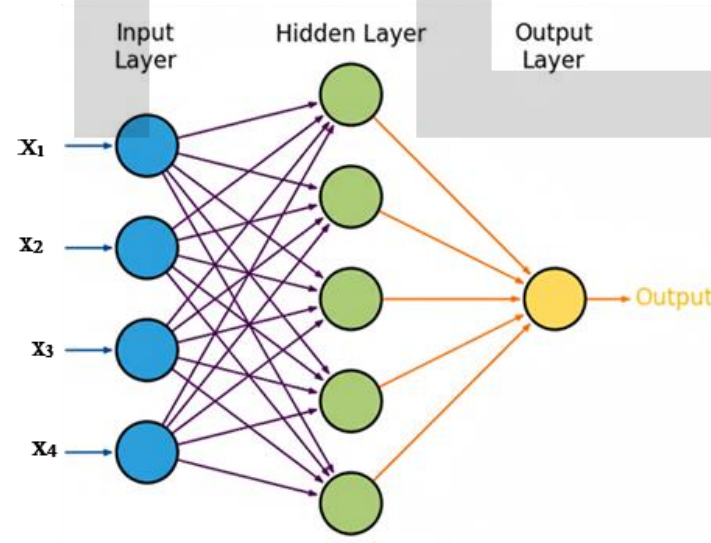
✓ The meaning of a word depends on its preceding words
(short-term dependency)

✓ Answer: French

Requires remembering earlier information.(Long-term dependency)

Can a feed-forward network work in this scenario?

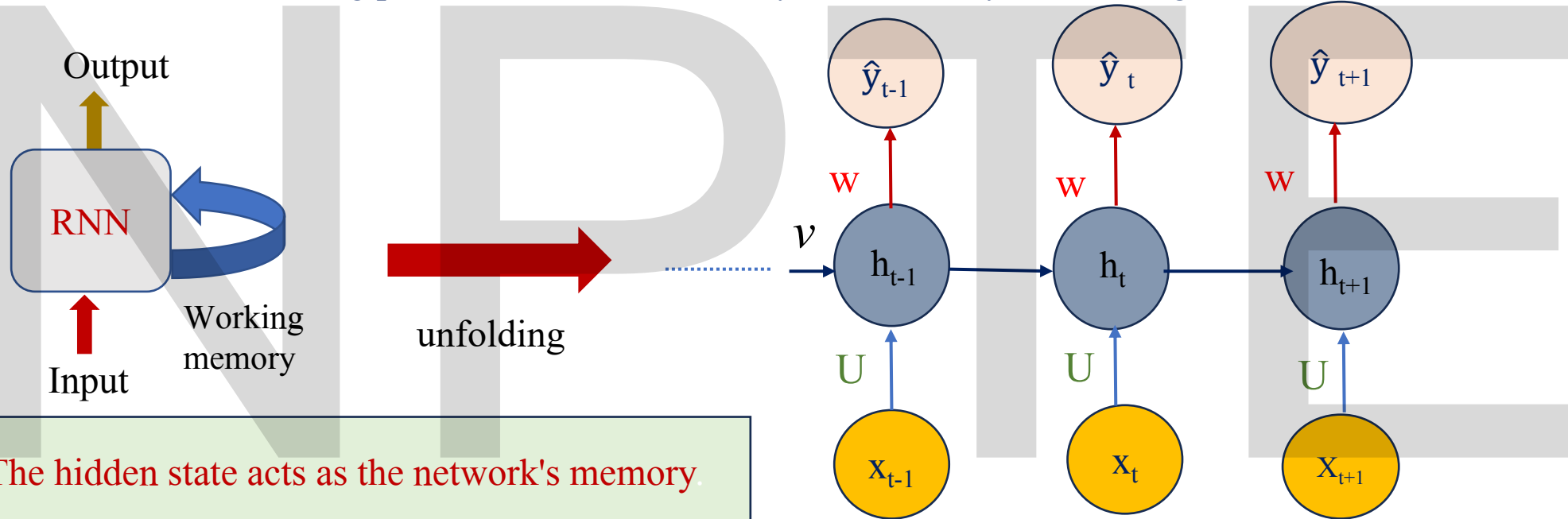
Feed-forward networks have no
memory of previous inputs.



Introduction to Recurrent Neural Networks

Core Idea

Instead of discarding previous information, carry it forward by introducing a hidden state



Weight matrices

U: Input $\rightarrow$ Hidden

V: Previous Hidden $\rightarrow$ Current Hidden (Recurrent weights)

W: Hidden $\rightarrow$ Output

Understanding RNN

Hidden state Update

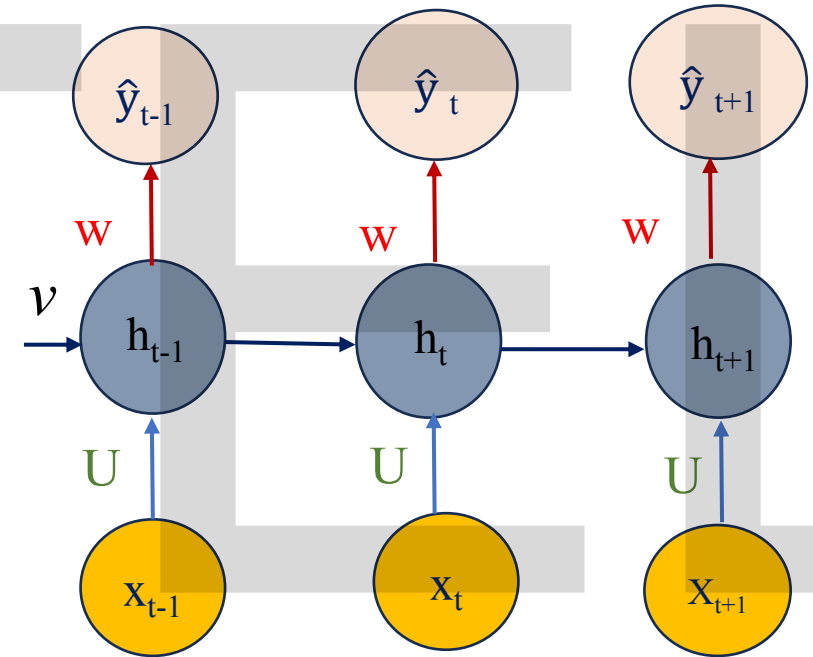
At each time step t , the RNN updates its hidden state using the current input x_t and the previous hidden state h_{t-1}

$$h_t = \tanh(vh_{t-1} + Ux_t + b_h)$$

Output Calculation

Output y_t at each step is calculated using the hidden state h_t (which depends on the input x_t)

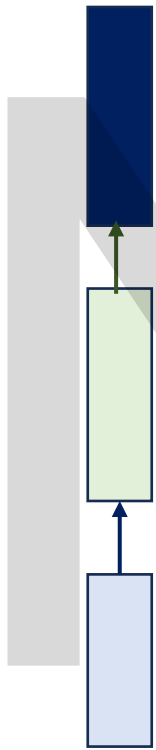
$$\hat{y}_t = g(wh_t + b_y)$$



The same weight matrices U , V , and W are shared across all time steps.

Types of RNN

One to One



Binary Classification

One to Many

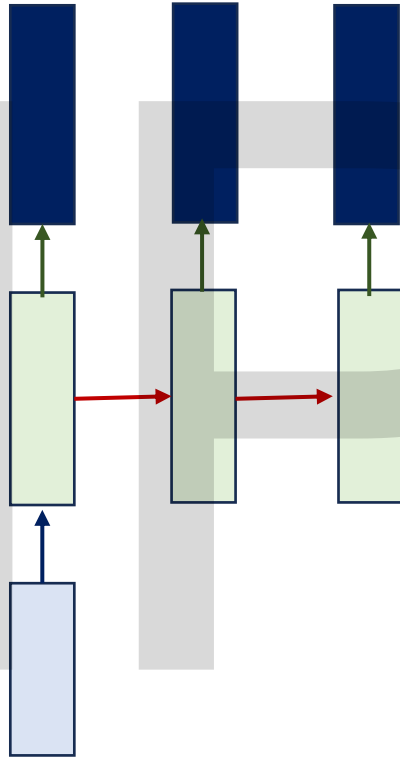
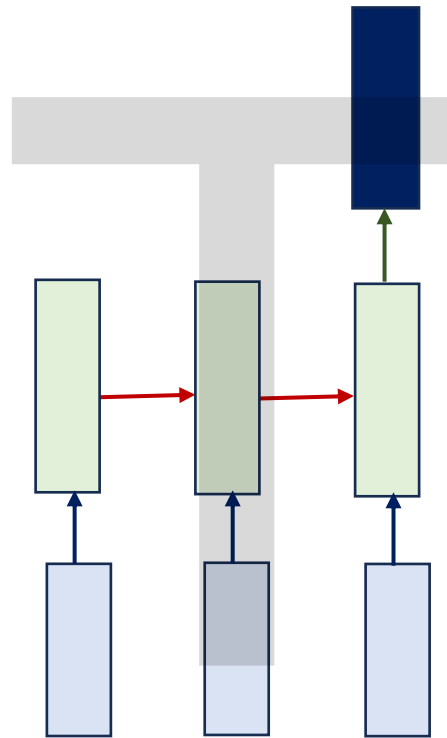


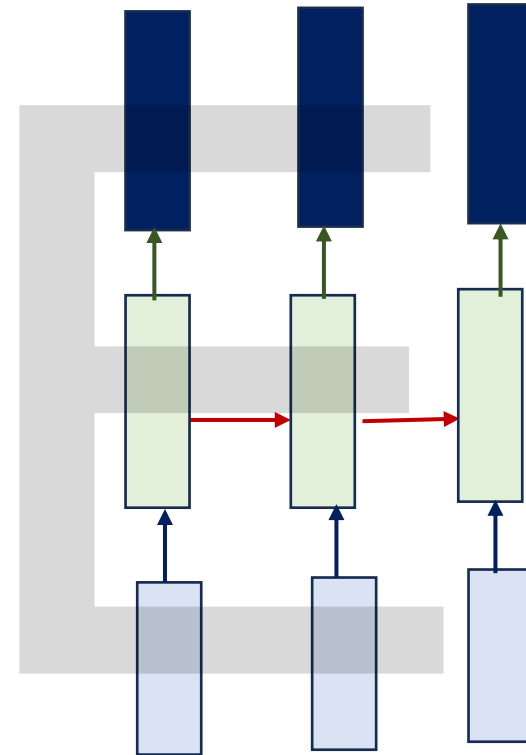
Image Captioning

Many to One



Sentiment Analysis

Many to Many



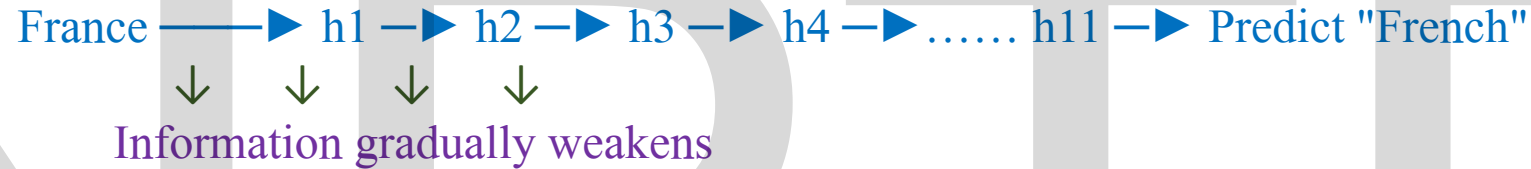
- Language translation
- Sequence labelling (POS tagging)



Limitation of RNNs

“My brother live in **France** and I visit him every summer, which is why I understand **French** quite well”

The connection between the location and the language is separated by many words.



RNNs cannot handle long-term dependencies

Why?

Vanishing Gradient

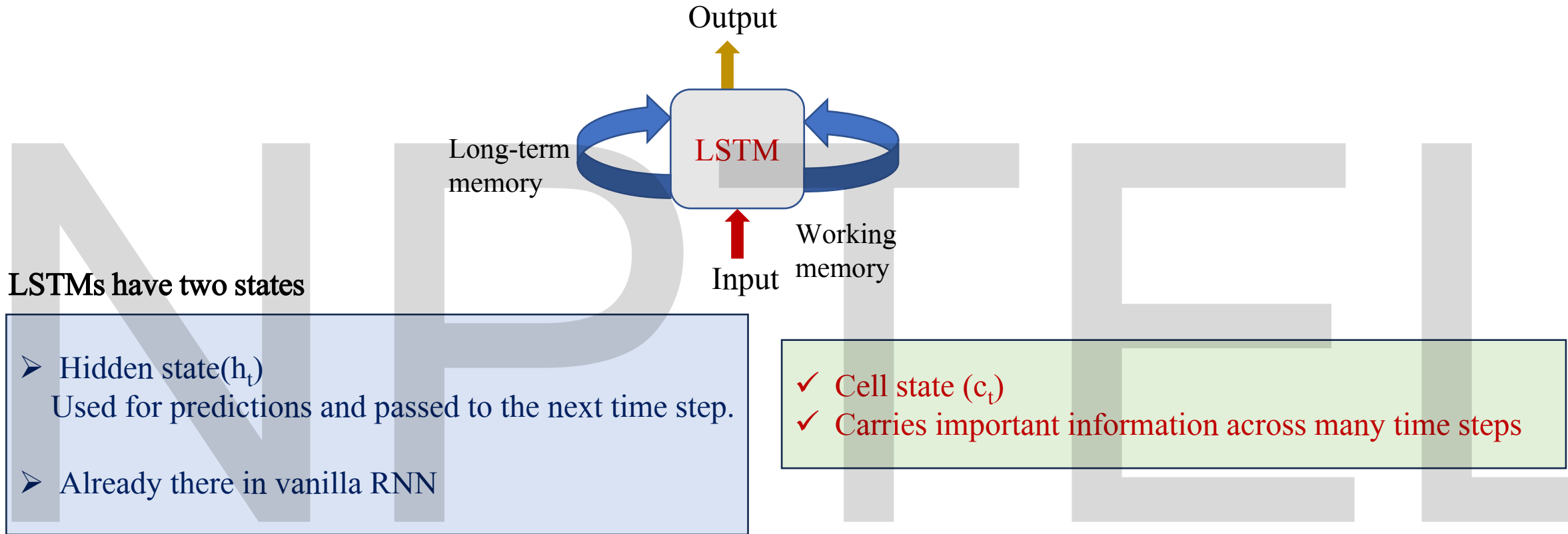
✓ Gradients vanish over long sequences, making it difficult to learn long-term dependencies.

Exploding Gradient

✓ When weights are large, the gradient explodes to infinity, making the network unstable.

Long Short term Memory

Long Short-Term Memory (LSTM)



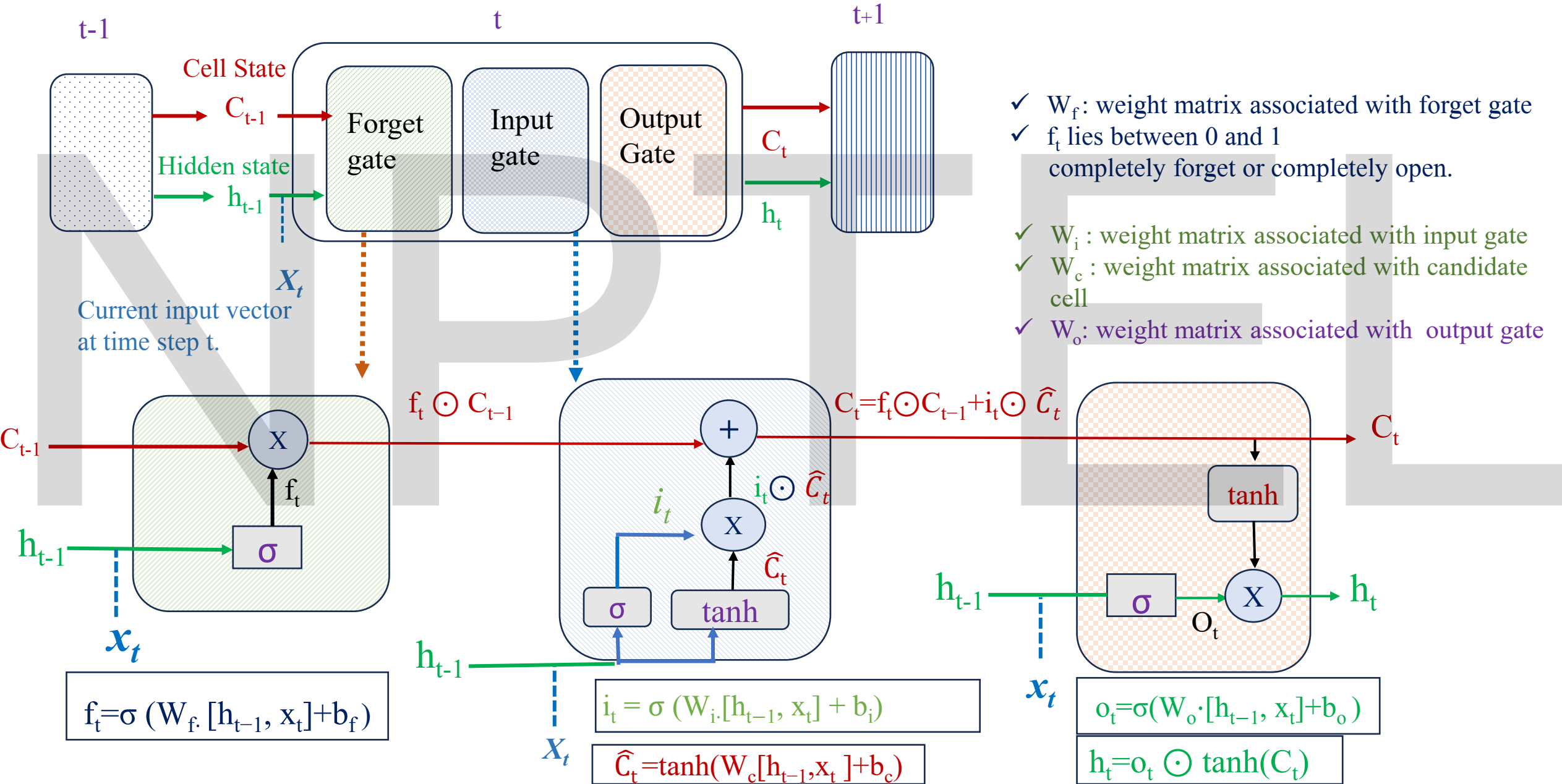
Gates in LSTM

- Forget Gate
- Input Gate
- Output Gate

- ✓ Gates are also neural networks with learnable functions
- ✓ Each gate learns how much information to retain, update, or expose.

LSTM Architecture

LSTM Chain with three nodes



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Summary

Next Session

Evolution of LSTMs to Transformers